

Nameer Iqbal Ansari

RTL Engineer | ASIC | RISC-V

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PROFESSIONAL SUMMARY

RTL Design Engineer (FPGA/ASIC) at XCLR Technologies, specializing in **RISC-V-based SoC** development. Certified **RISC-V Foundational Associate (RVFA)** and **Google Summer of Code (GSoc) 2022 contributor** and **2023 mentor**, with hands-on experience in **SystemVerilog/Verilog** based **ASIC** and **FPGA** designs, **Lint/CDC/RDC** sign-off, **timing closure**, and **verification**.

WORK EXPERIENCE

02/2025 – Present

RTL Design Engineer (FPGA/ASIC), XCLR Technologies (Pvt.) Ltd.

- **High-Speed I/O Subsystem (HSIOSS)** — Led design and verification across **3 teams** and **6 engineers** (excluding IP vendor teams) for a subsystem integrating a **32 Gbps** SerDes and **PCIe Gen5** controller, running at a **1.5 GHz** core clock with a **100 MHz** reference clock and **2048-bit data / 64-bit** configuration **AXI interfaces**. PIPE interface between **MAC** and **PHY** spanning **8 PCIe lanes**, each interfacing through a **32-bit** PIPE datapath. Defined the verification test plan including directed chip-level firmware testing alongside block-level SystemVerilog testbenches. Also utilized and modified the existing UVM testbenches within the IPs
- **RISC-V Vector Load/Store Unit (VLSU)** — Designed a **2048-bit** RVV load/store unit interfacing to a **64-bit** memory port, with **AXI-ID tracking**, address generation and translation, and data packing/unpacking logic supporting configurable element widths (**8–64 bits**) within a single vector register. Added multi-stage **pipelining** to **reduce latency** and **cell count**
- **IP Replacement Across 5 Subsystems** — Re-architected 5 subsystems by replacing Synopsys-sourced data/protocol converters, crossbars, **CDC** synchronizers, and **AXI-to-native-memory bridges** with PULP Platform IP. Utilized Synopsys **CoreAssembler** to generate a GUI-based integrated RTL, and also used **CoreConsultant** to generate standalone Synopsys IPs
- **Unified DFT/Functional JTAG Architecture** — Architected a single chip-level JTAG interface for two independent TAP controllers (**DFT scan-based** and functional **RISC-V PULP debug**) by **muxing** in between both TAP interfaces onto one physical port, enabling both production test and hardware debug through shared JTAG standard pins
- **Control-Register-to-AXI Bridge** — Designed and verified a protocol bridge between a **16-bit** native SerDes control-register interface and a **64-bit** AXI configuration bus, resolving **CDC** and 64-to-16-bit **data-width conversion**. Validated the design with **block-level SV TB** and **system-level firmware** testing after integration of the converter into the chip
- **512-bit eFuse Controller** — Implemented **32-bit AXI** access by bridging the native **eFuse** memory array interface to an **AXI-based** register-access architecture, including command-driven read/write transaction handling. Performed SystemVerilog-based block-level testing and validated end-to-end chip-level functionality through **QSPIS** interface access testing
- **Chip-level low-power implementation** — Performed PPA analysis and enabled up to **80%** of the chip to operate in low-power mode using **low-power** features provided by Synopsys IPs and custom design logic. Created a **safe power-switch-off** mechanism that ensured **glitch-free clock gating** for low-power entry and in-flight **AXI** transaction handling by safely handling incomplete or unintended AXI transactions via AXI response error (**SLVERR/DECERR**) functionality during power-state transitions

09/2023 – 02/2025

Associate Engineer, Xcelerium

- **12nm timing closure at 1.5 GHz of RISC-V-based core subsystem** — Achieved **-0.01** worst negative slack (**WNS**) and reduced setup and hold (**reg2out/in2reg/in2out**) violations across **500+** violating paths by optimizing combinational logic, skew balancing, registering, floorplanning, and SDC constraint fixes
- **JTAG daisy Chain at Aries** — Designed and validated a **daisy-chained JTAG** debug architecture linking two PULP **riscv-dbg** TAP controllers across two subsystems, allowing either core to be halted and debugged independently while the other ran normally. Verified via **SimJTAG (C DPI)** in simulation and on **AWS EC2 cloud FPGA**
- **Lint, CDC, and RDC sign-off** — Accomplished **0 error/warning** sign-off across 4 subsystems, each containing a minimum of **100** violations that are detected using **Synopsys SpyGlass**, ensuring digital designs are ready for physical design (PD)
- **RTL design & Digital IP integration** — Integrated digital components in the RTL designs including logic analyzers, **CDC** synchronizers, **AXI** to **APB** protocol converters, data width converters, crossbar interconnects, register files and **FSMs**
- **Technical Documentation & Design Reviews** — Authored comprehensive design documents, verification test plans, timing diagrams, and block diagrams to aid design reviews and knowledge transfer

03/2020 – 09/2023

Research Assistant, Micro Electronics Research Lab (MERL)

- **RV32I Core Design** — Built an open-source **RV32I CPU core** [🔗](#) starting from scratch using SystemVerilog, simulated with **Verilator** using **FuseSoC**, and emulated on **Arty A7 FPGA** using **Xilinx Vivado**. Later integrated basic **UVM** and **JasperGold** support
- **Out-of-Order Upgrade to RV32I Core** — Extending the **RV32I** core with out-of-order execution capability, adding 2-wide instruction fetch, register renaming via a **Register Alias Table (RAT)**, a reorder buffer (**ROB**) for in-order commit, and reservation stations feeding ALU and branch execution units. Migrating the existing in-order pipeline to a **dispatch/issue/commit** structure inspired by **Berkeley's BOOM** design

- **NOVA1 SoC Development** — Contributed to **NOVA1 System-on-Chip (SoC)** development and its **AWS EC2-based** emulation, and **Zyper OS** bring-up
- **Multicore Cache Coherency & CHI Research** — Led the research of the **multicore architecture** and its **cache coherency** through the **OpenPiton** framework. Also researched converting OpenPiton's native coherence protocol to ARM's **CHI**, mapping its directory-based coherence engine onto CHI's **RN-F/HN-F/SN-F** node abstractions
- **RISC-V Training & Mentorship** — Conducted **student training programs** on RTL design and how to contribute to open-source
- **NOVA1 Presentation** — Presented NOVA1 project at the **RISC-V Karachi Meetup**

TECHNICAL SKILLS

- **RTL Design** — RISC-V VLSU | JTAG | SerDes | PCIe | eFuse | UART | CDC | XBAR | Cache | AXI | APB | TileLink
- **EDA Tools** — VCS | Verdi | Xilinx Vivado | GTKWave | CoreAssembler | CoreConsultant | QuestaSim | Verilator | FusionCompiler | PrimeTime
- **Formal Verification** — JasperGold | SVA (SystemVerilog Assertions)
- **Static Analysis** — Synopsys SpyGlass (Lint,CDC,RDC)
- **Languages** — Verilog | SystemVerilog | SystemVerilog-OOP | UVM (basic) | C | C++ | RISC-V Assembly
- **Scripting** — Tcl | Shell | Makefile | CI/CD | Python | YAML (FuseSoC)
- **Version Control** — GitHub | GitLab
- **Operating System** — Linux | Windows

SOFT SKILLS

- **Languages** — English (C1), Urdu (Native)
- **Other** — Design review | Technical Documentation | Debugging & Root-Cause Analysis | Design verification coordination | Technical Leadership | Timing-Closure Problem Solving | Research & Mentorship | Volunteer work

PROJECTS

Google Summer of Code (GSoC), Contributor (2022) / Mentor (2023)

- Added **FuseSoC** support on the **Azadi-SoC** and **rebased** the **lbex core** to latest upstream commits ([GSoC'22](#)) [↗](#)
- Mentored **TileLink** Uncached Heavyweight (**TL-UH**) implementation in **Azadi-SoC** ([GSoC'23](#)) [↗](#)

MARCore & NOVA1, Research Lead / Contributor

- Researched multicore cache-coherent memory systems using OpenPiton workflow and Ariane Core ([MARCore1](#)) [↗](#)
- Contributed to NOVA1 SoC that contains RV64GC Core, UART, BRAM, DDR, and JTAG support ([The NOVA Project](#)) [↗](#)

EDUCATION

2019 – 2023

Bachelor of Engineering in Electrical Engineering, Usman Institute of Technology

(CGPA: 3.289) (Specialization: Computer Systems)

- **Major Courses** — Digital Logic Design | Computer Architecture | Microprocessor-Based Systems | Embedded Systems | Operating Systems

Honors & Awards

- Hasham Foundation Scholarships by Usman Institute of Technology in 2022
- Linux Foundation (LiFT) Scholarship Recipient in 2024

CERTIFICATES

- **RISC-V Foundational Associate (RVFA)** — Date: Aug 2025 | Issuer: RISC-V International | Sponsor: Linux Foundation | [Link](#) [↗](#)
- **RISC-V Fundamental (LFD210)** — Date: Aug 2025 | Issuer & Sponsor: Linux Foundation | [Link](#) [↗](#)

PUBLICATIONS & ARTICLES

- Multicore SoC For SMP Linux ([Thesis](#)) [↗](#)
- NOVA Emulation Project- An open-source RISC-V-based application-class AI SoC ([Project Doc](#)) [↗](#)
- A Summer of Opportunity ([Medium — GSoC experience](#)) [↗](#)
- Abstractions Can Lead Humanity to Ignorance (An Unpopular Opinion) ([Medium — technical opinion](#)) [↗](#)